Predicate Calculus

by

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First recall the concept of a function $f : A \to B$. For all $x \in A$, $f(x) \in B$. Now, let A be a non-empty set and S be the set of all **propositions**. Then the function $P : A \to S$ is called a **Propositional Function**. So for any $x \in A$, $P(x) \in S$, i.e., $P(x)$ is a proposition.

Here the set A is called the **Domain** of the propositional function P. $P(x)$ is called the **Predicate** and x is called the **Predicate Variable** of the propositional function P.

Example: (1) Let $P : \mathbb{R} \to S$ be a propositional function, defined as $P(x) : x > 25$, where $\mathbb{R}$ be the set of all real numbers and S is the set of all **propositions**. This means "*for every* $x \in \mathbb{R}$, $x > 25$" is a proposition, 'x' is predicate variable and '$x > 25$' is predicate and 'P' is the predicate function. For example, $P(10) : 10 > 25$ is a proposition (which is False), and $P(28.5) : 28.5 > 25$ is a proposition (which is True).

Here, for every $x \in \mathbb{R}$ we get, a $P(x) \in S$.

(2) Let, $P((x, y)) : 3x + 2y = 1$ where x, y are real numbers. Here the mapping $P : \mathbb{R} \times \mathbb{R} \to S$ defined by $P((x, y)) : 3x + 2y = 1$ is a propositional function. $3x + 2y = 1$ is the **predicate**, and (x, y) is the **predicate variable**.

Quantifications & Quantifiers

In logic, Quantification is a way of creating a proposition from a propositional function, using suitable symbols, and those symbols are called quantifiers. There are mainly two types of quantifiers.

Universal Quantifier

The phrase *"for all"*, denoted by $\forall$, is called the **universal quantifier**.

For example, consider the sentence *"All human beings are mortal"*.

Let U be the domain set of *"all human beings"*. Consider the propositional function $P(x)$: *"x is mortal."* Then the above sentence can be written as

$$(\forall x \in U)P(x) \quad \text{or simply} \quad \forall x\, P(x).$$

The expression $P(x)$ by itself is an open sentence and hence it has no truth value. However, "$\forall x\, P(x)$" does have a truth value.

The truth values of "$\forall x\, P(x)$" are assigned as follows:

- "$\forall x\, P(x)$" is **true** if and only if $P(x)$ is true for every x in U, where U is the domain of x.
- "$\forall x\, P(x)$" is **false** if and only if $P(x)$ is false for at least one $x \in U$.

Existential Quantifiers

The phrase *"there exists"*, denoted by $\exists$, is called the **existential quantifier**.

For example, consider the statement *"There exists a real number x such that $x^3 = 2$"*.

Let $\mathbb{R}$ be the set of all real numbers. Then the above sentence can be expressed as

$$(\exists x \in \mathbb{R})P(x) \quad \text{or simply} \quad \exists x\, P(x)$$

where $P(x)$: *"$x^3 = 2$"*.

The expression $\exists x\, P(x)$ has truth value and, as such, it is a statement and is called an **existential statement**.

The truth values of $\exists x\, P(x)$ are assigned as follows:

- $\exists x\, P(x)$ is **true** if and only if $P(x)$ is true for at least one $x \in U$, where U is the domain or universe of x.
- $\exists x\, P(x)$ is **false** if and only if $P(x)$ is false for every $x \in U$.

For example, consider the statement: *"There is a student of B.Tech. who has taken the course of discrete mathematics."*

This statement can be written as $(\exists x)P(x)$ where the domain of x is the set of all students of B.Tech and P is the predicate *"has taken the course of discrete mathematics"*.

The negation of the above statement is *"It is not the case that there is a student of B.Tech who has taken the course of discrete mathematics."*

Or, equivalently,

"All students of B.Tech have not taken the course of discrete mathematics."

Hence the negation of $(\exists x)P(x)$ is $(\forall x) \sim P(x)$, i.e.,

$$\sim ((\exists x)P(x)) \equiv (\forall x) \sim P(x),$$

Table 1: Equivalences of quantified statements and their negations

Quantified Statements	**Negated Quantified Statements**
$(\forall x)P(x)$	$(\exists x) \sim P(x)$
(all are true)	(at least one is false)
$(\forall x) \sim P(x)$	$(\exists x)P(x)$
(all are false)	(at least one is true)
$(\exists x)P(x)$	$(\forall x) \sim P(x)$
(at least one is true)	(all are false)
$(\exists x) \sim P(x)$	$(\forall x)P(x)$
(at least one is false)	(all are true)